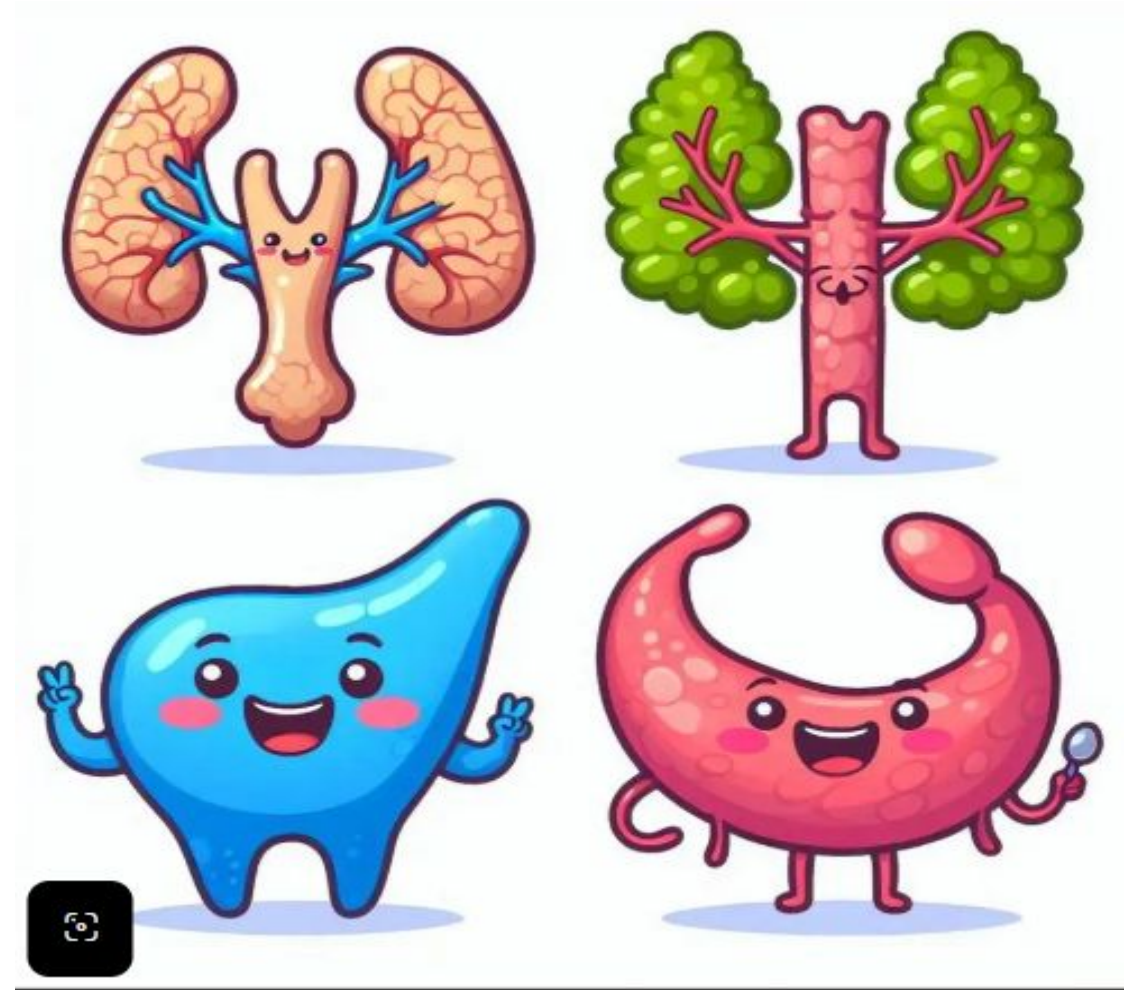


ENDOCRINE REGULATION

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DIKIOHS,DUHS



LEARNING OBJECTIVES:

- Feedback mechanisms: negative and positive feedback.
- Hormonal regulation and its role in body functions.
- Hormones and their actions.
- Relationship between feedback mechanisms, hormonal regulation, and homeostasis.
- Role of the endocrine system in maintaining internal balance.

INTRODUCTION

ENDOCRINE GLANDS:

- Endocrine glands are the glands which synthesize and release the classical hormones into the blood.
- Endocrine glands are also called ductless glands because the hormones secreted by them are released directly into blood without any duct. E.g: pituitary gland, ovary and testis, thyroid and parathyroid glands, pancreas and adrenal glands.
- Endocrine glands play an important role in homeostasis and control of various other activities in the body through their hormones.
- The mediators of such control mechanisms are soluble factors known as hormones.

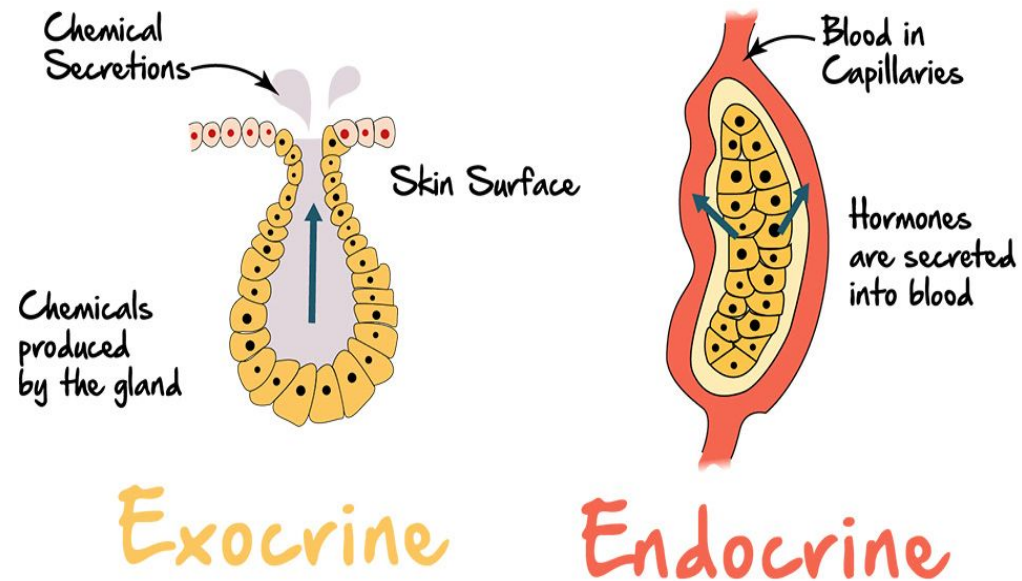
Major endocrine glands;

- Hormones secreted by endocrine glands
- Hormones secreted by gonads
- Hormones secreted by other organs
- Local hormones

EXOCRINE GLANDS

- Exocrine glands secrete their products into ducts that carry the secretion to a specific location (skin, digestive tract).
- Examples: Salivary glands (salivary amylase in oral cavity), mammary glands, digestive glands like pancreas (exocrine part) and liver (via bile ducts).

Endocrine vs Exocrine Glands



HORMONES AND THEIR ACTION

Hormones are chemical messengers, synthesized by endocrine glands. Based on chemical nature, hormones are classified into three types

1. Steroid hormones
2. Protein hormones
3. Derivatives of the amino acid called tyrosine

CLASSIFICATION OF HORMONES DEPENDING UPON CHEMICAL NATURE

STEROID HORMONES:

Steroid hormones are the hormones synthesized from cholesterol or its derivatives. They are lipid soluble hydrophobic molecules. Steroid hormones are secreted by adrenal cortex, gonads and placenta.

E.g: aldosterone, cortisol, estrogen, testosterone and progesterone.

PROTEIN HORMONES:

Protein hormones are large or small peptides. They are water soluble. They are secreted by pituitary gland, parathyroid glands, pancreas and placenta ('P's).

E.g: oxytocin, vasopressin, calcitonin, insulin.

AMINES HORMONES:

They are derived from the amino acid tyrosine. They are secreted from the thyroid and the adrenal medulla.

Steroids	Proteins	Derivatives of tyrosine
Aldosterone	Growth hormone (GH)	Thyroxine (T_4)
11-deoxycorticosterone	Thyroid-stimulating hormone (TSH)	Triiodothyronine (T_3)
Cortisol	Adrenocorticotrophic hormone (ACTH)	Adrenaline (Epinephrine)
Corticosterone	Follicle-stimulating hormone (FSH)	Noradrenaline (Norepinephrine)
Testosterone	Luteinizing hormone (LH)	Dopamine.
Dihydrotestosterone	Prolactin	
Dehydroepiandrosterone	Antidiuretic hormone (ADH)	
Androstenedione	Oxytocin	
Estrogen	Parathormone	
Progesterone	Calcitonin	
	Insulin	
	Glucagon	
	Somatostatin	
	Pancreatic polypeptide	
	Human chorionic gonadotropin (HCG)	
	Human chorionic somatomammotropin.	

CONTROL OF HORMONE RELEASE

- **Hormonal stimuli:** This occurs when one hormone stimulates the release of another hormone. For example, the anterior pituitary releases thyroid-stimulating hormone (TSH), which stimulates the thyroid gland to produce T3 and T4.
- **Humoral stimuli:** These are responses to changes in extracellular fluids, such as blood glucose levels, which trigger the release of hormones like insulin from the pancreas.
 - High blood glucose: Insulin secretion
 - Low blood glucose: Glucagon secretion
- **Neural stimuli:** This involves direct stimulation of endocrine glands by the nervous system, such as during stress responses, where hormones like epinephrine and norepinephrine are released from adrenal medulla.

These mechanisms work together to regulate hormone levels and maintain homeostasis in the body.

FEATURES OF HORMONES

- Hormones are low molecular weight chemical messengers.
- They are secreted in trace amounts in response to specific secretory stimuli.
- They are soluble in water and act as catalyst
- A single hormone have multiple effects on a single target tissue or on several different tissues.
- Hormones have high degree of action of specificity.
- They are inactivated as soon as their functions are over.
- Endocrine glands are under the control of nerves.

HORMONES RECEPTORS

Hormones receptors are big protein molecules found in target cells. There are thousands of receptors on each cell but each receptor recognizes only one hormones.

Examples of hormones receptors:

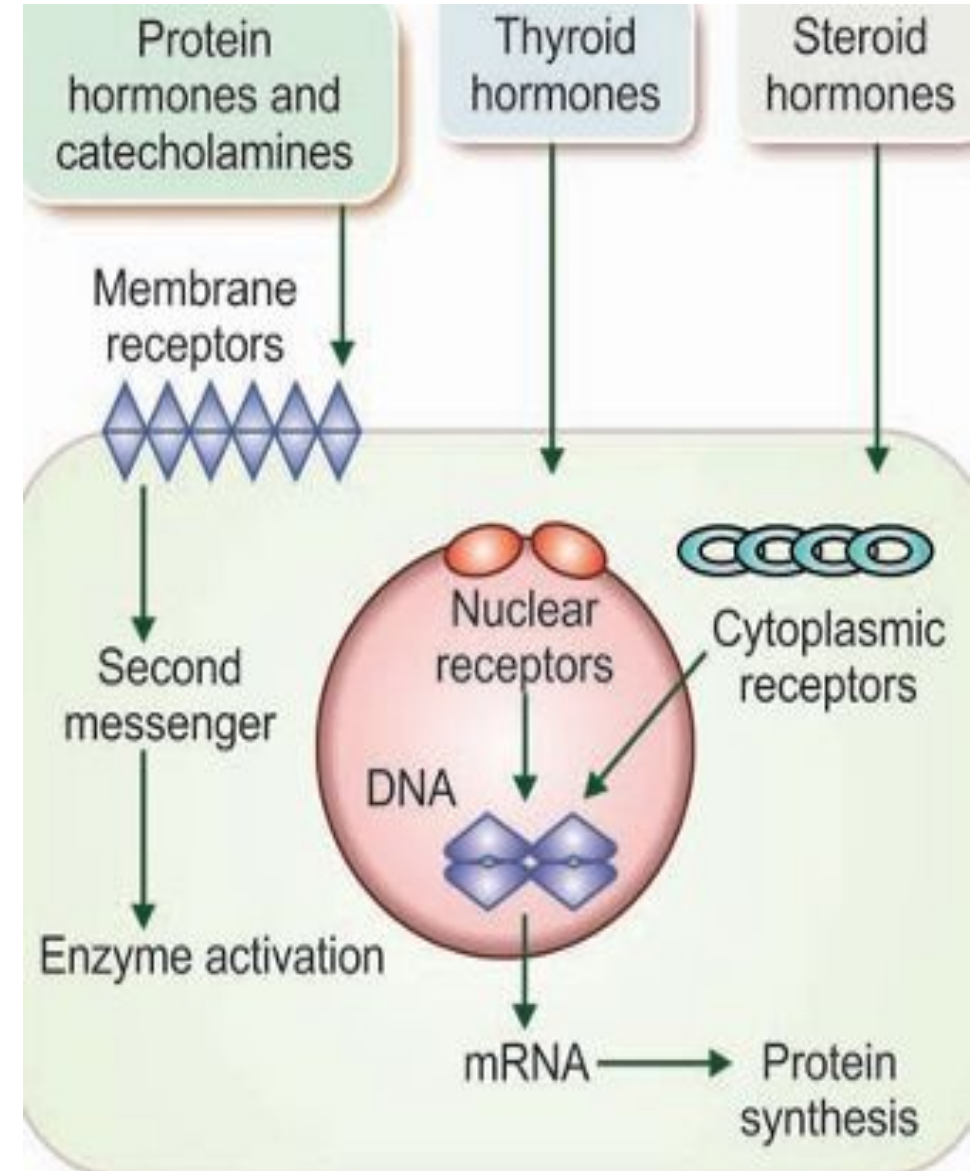
- Glucagon receptors
- Androgen receptors
- Insulin receptors
- Progesterone receptors

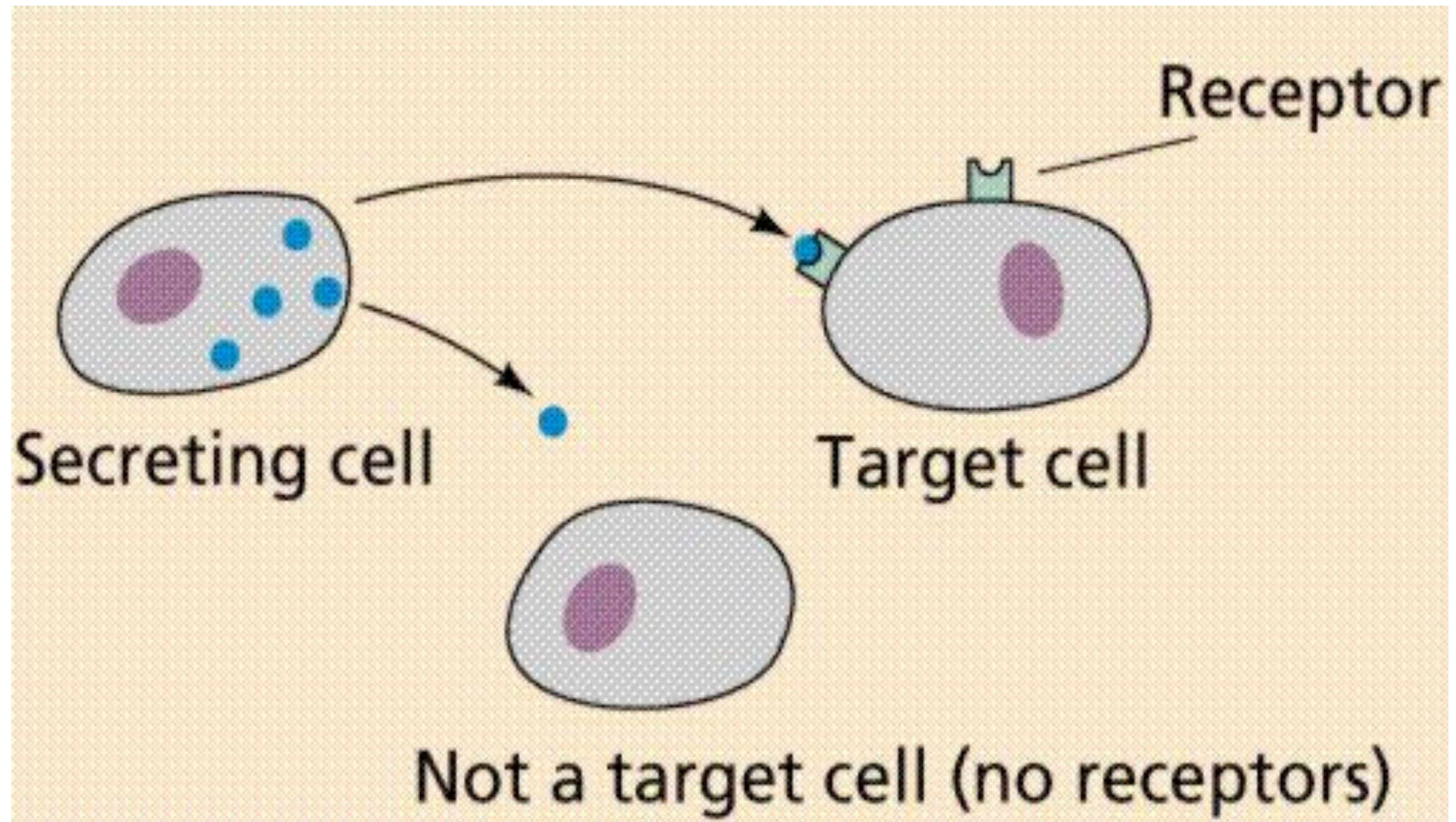
These hormones receptors enable the transmission of signals and also aid in regulation of gene expressions.

SITUATION OF THE HORMONE RECEPTORS

Hormone receptors are situated either in cell membrane or cytoplasm or nucleus of the target cells as follows:

1. Cell membrane: Receptors of protein hormones and adrenal medullary hormones (catecholamines) are situated in the cell membrane.
2. Cytoplasm: Receptors of steroid hormones are situated in the cytoplasm of target cells
3. Nucleus: Receptors of thyroid hormones are in the nucleus of the cell.





Regulation of Hormone Receptors

The number and affinity of receptor may change

DOWN REGULATION: when a hormone is secreted in excess, the number of receptors of that hormone decreases due to binding of hormone with receptors. This process is called down regulation.

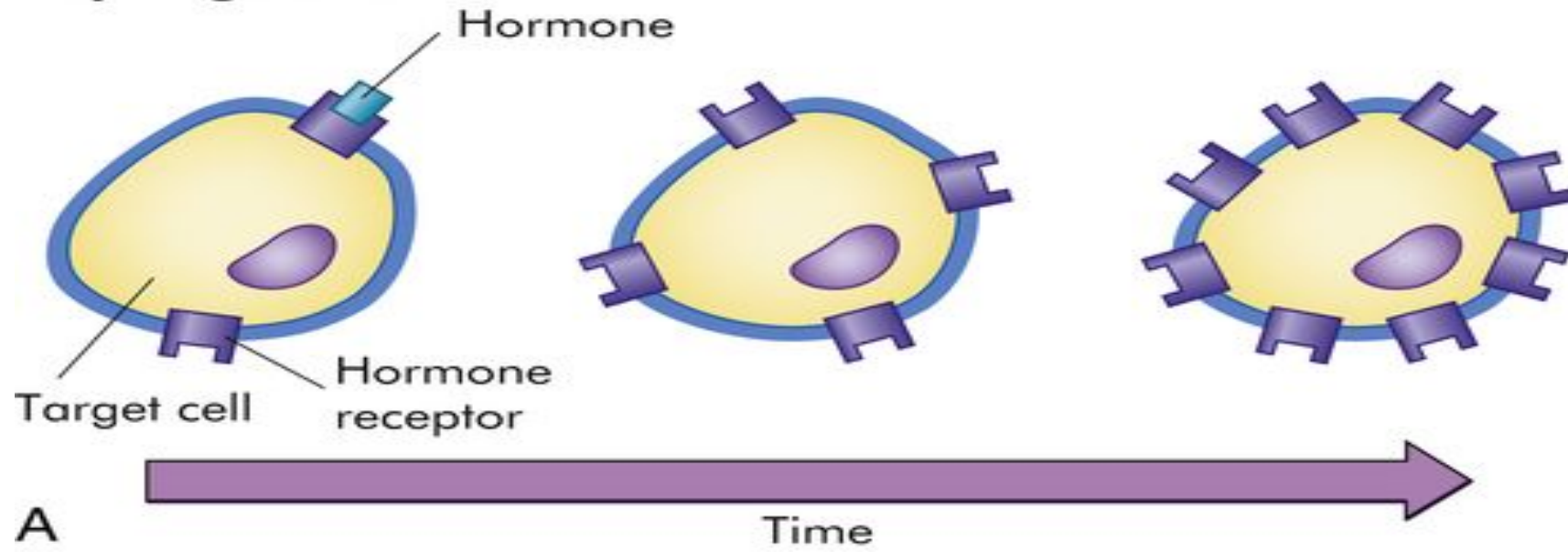
-occurs after long exposure to high level of polypeptide hormones e.g adipose cells to insulin.

UP REGULATION: During the deficiency of the hormone, the number of receptor increases, which is called upregulation.

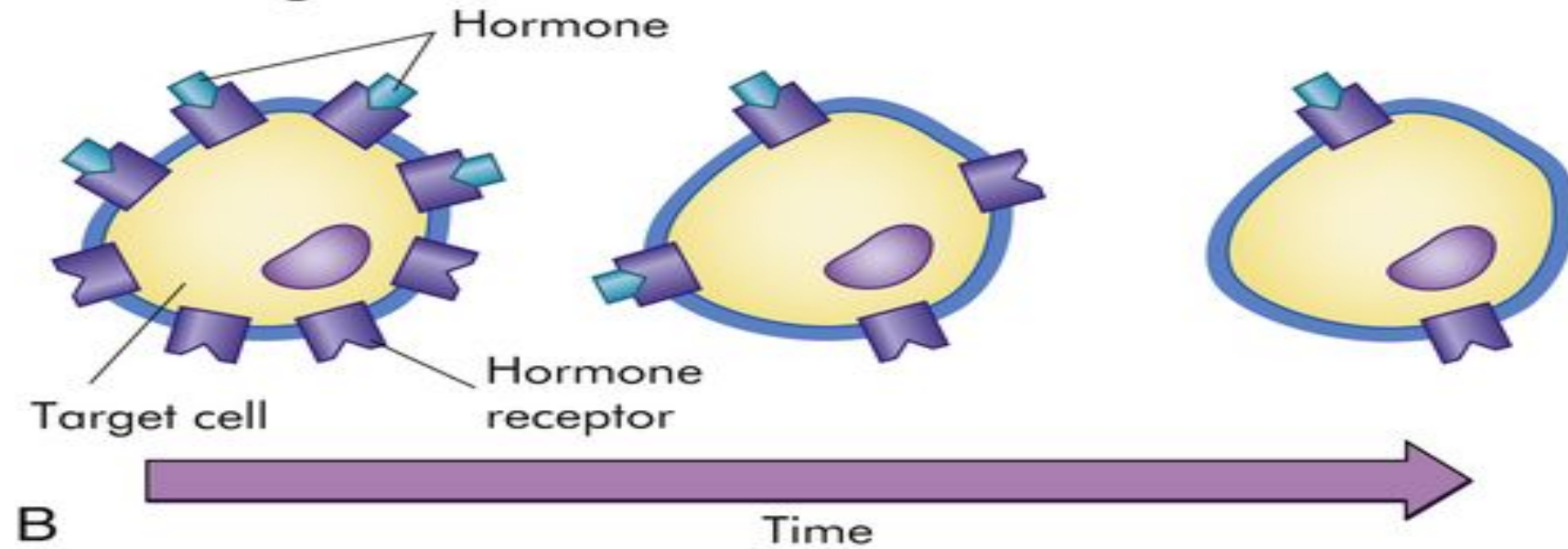
E.g: GnRH secreted by the hypothalamus increases sensitivity of anterior pituitary cells to further stimulation (upregulation of receptors).

INTERNALIZATION: Hormone in the form of hormone-receptor complex enters the target cell by means of endocytosis and executes the actions. The whole process is called internalization.

Up-regulation



Down-regulation

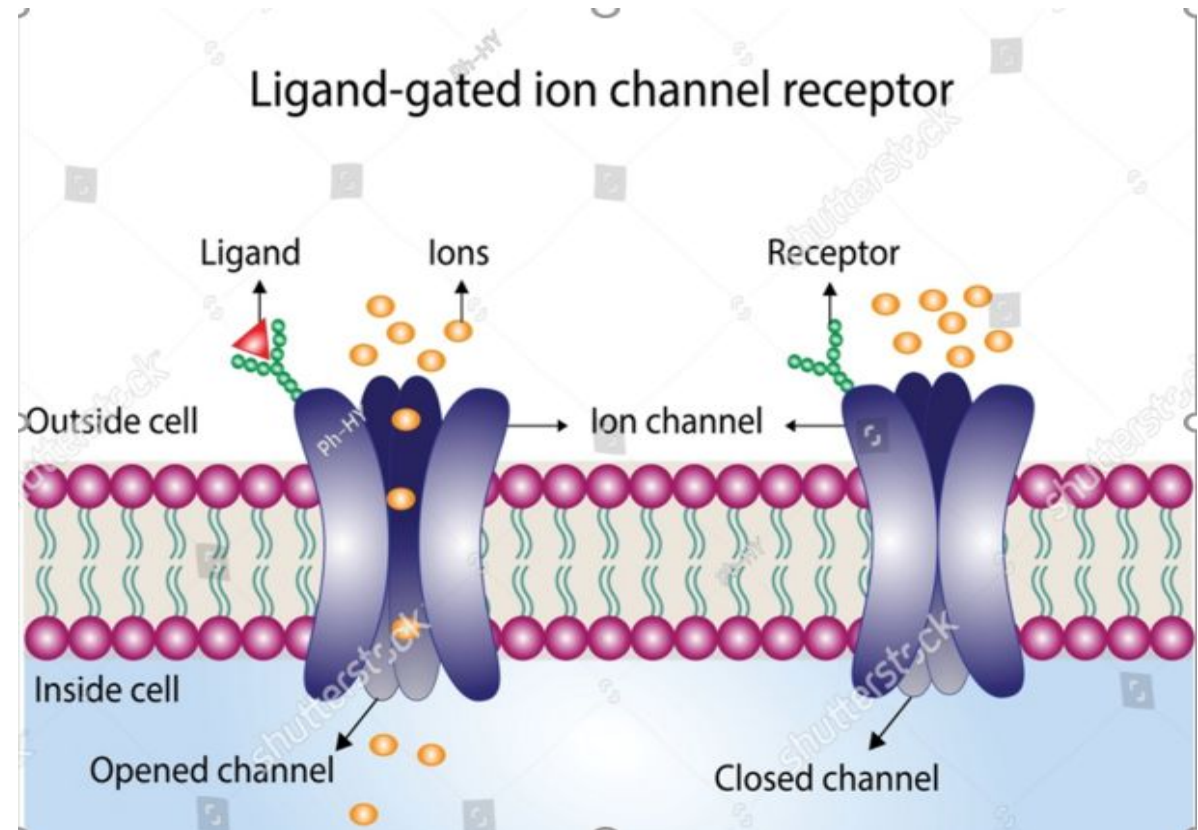


MECHANISM OF HORMONAL ACTION

Hormone does not act on the target cell directly. It combines with receptor to form hormone-receptor complex.

1:ION CHANNEL LINKED HORMONE RECEPTORS (MEMBRANE PERMEABILITY MECHANISM):

Neurotransmitters e.g norepinephrine combine with receptors –change in structure of receptors –opening or closing of a channel for Na, K ,or Ca- depolarization of postsynaptic membrane depending upon nature of ions.



2: BY ACTIVATING INTRACELLULAR ENZYME:

- Protein hormones and the catecholamines act by activating the intracellular enzymes.
- First Messenger: The hormone which acts on a target cell, is called first messenger or chemical mediator. It combines with the receptor and forms hormone-receptor complex.
- Second Messenger Hormone-receptor complex activates the enzymes of the cell and causes the formation of another substance called the second messenger or intracellular hormonal.
E.g: Cyclic AMP, cAMP

Sequence of events in the formation of cAMP

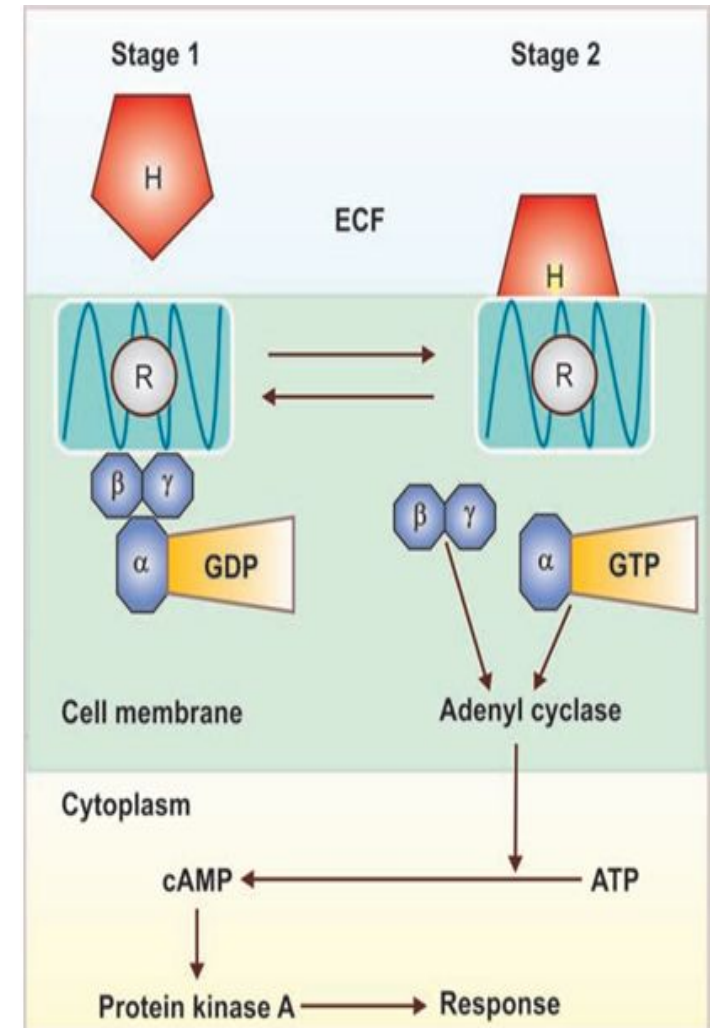
- i. Hormone binds with the receptor in the cell membrane and forms the hormone-receptor complex
 - ii. It activates the G protein
 - iii. G protein releases GDP from α -GDP unit
 - iv. The α -subunit now binds with a new molecule of GTP, i.e. the GDP is exchanged for GTP
 - v. This exchange triggers the dissociation of α -GTP unit and β - γ dimer from the receptor
 - vi. Both α -GTP unit and β - γ dimer now activate the second messenger pathways
 - vii. The α -GTP unit activates the enzyme adenylyl cyclase.
 - viii. Activated adenylyl cyclase converts the adenosine triphosphate of the cytoplasm into cyclic adenosine monophosphate (cAMP) .
- When the action is over, α -subunit hydrolyzes the attached GTP to GDP by its GTPase activity

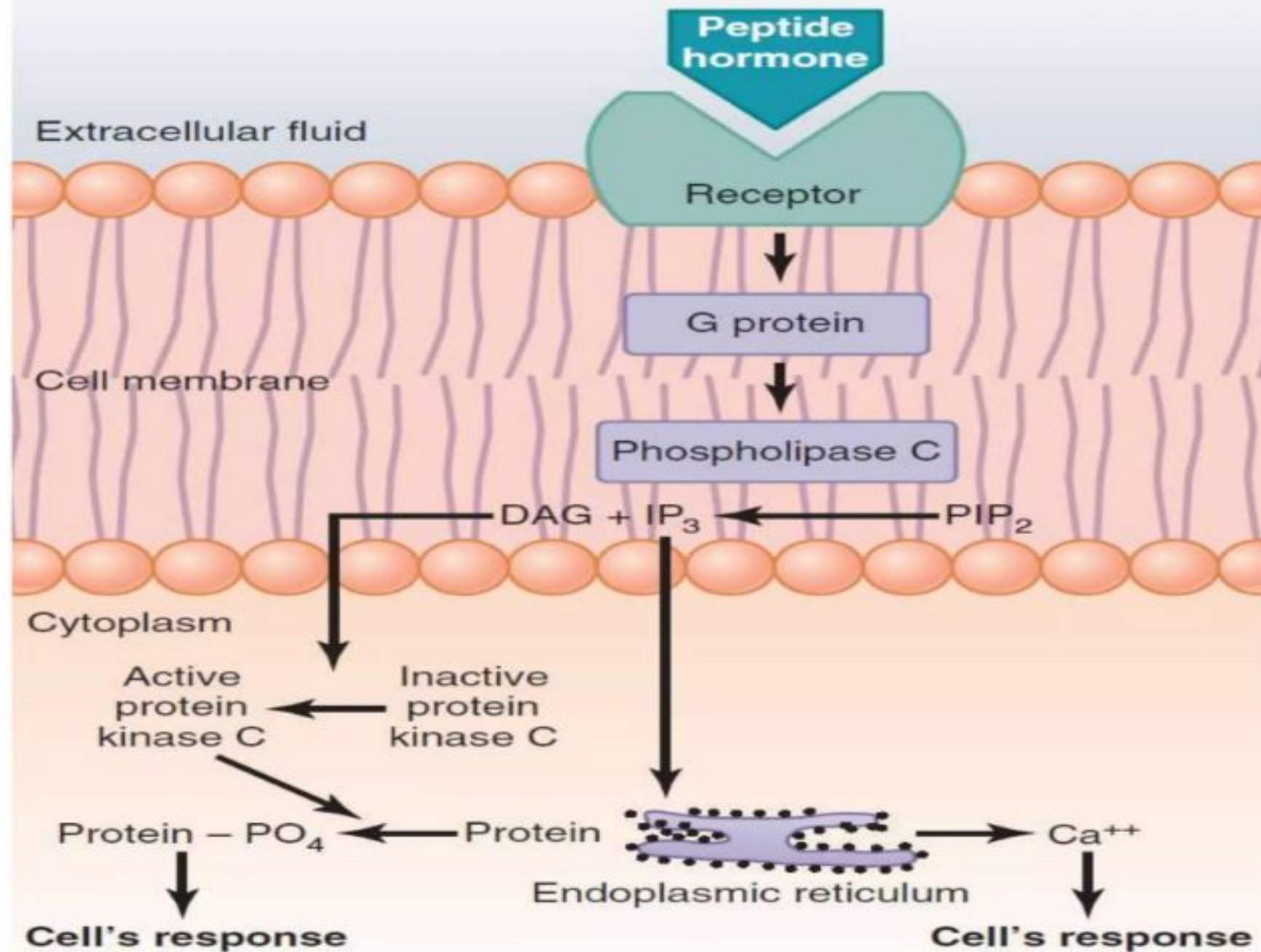
Cyclic AMP executes the actions of hormone inside the cell by stimulating the enzymes like protein kinase A. Cyclic AMP produces the response, depending upon the function of the target cells through these enzyme

- i. Contraction and relaxation of muscle fibers
- ii. Alteration in the permeability of cell membrane
- iii. Synthesis of substances inside the cell
- iv. Secretion or release of substances by target cell

Other Second Messengers:

- i. Calcium ions and calmodulin
- ii. Inositol triphosphate
- iii. Diacylglycerol (DAG)



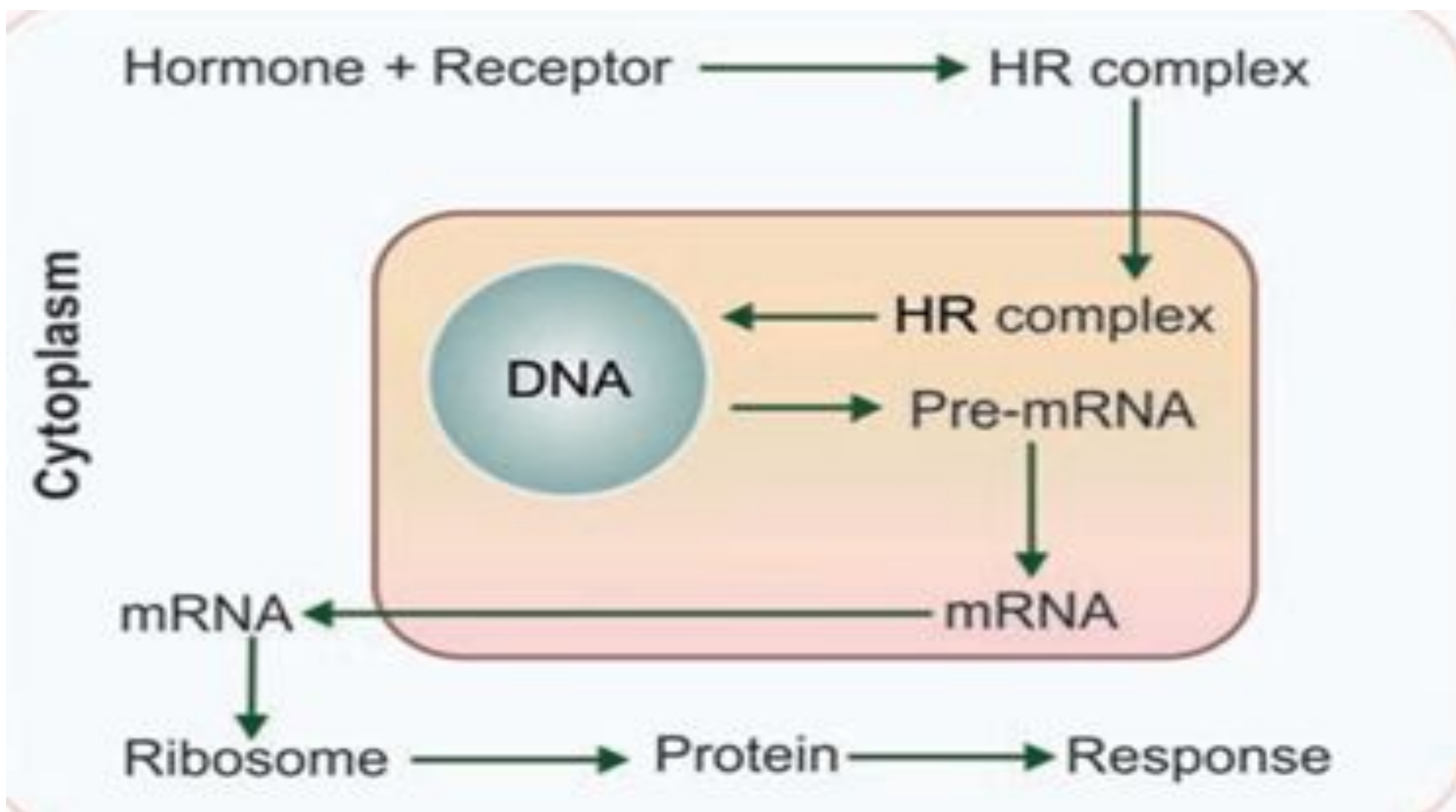


3: BY ACTING ON GENES

Thyroid and steroid hormones execute their function by acting on genes in the target cells.

Sequence of Events during Activation of Genes

- i. Hormone enters the interior of cell and binds with receptor in cytoplasm (steroid hormone) or in nucleus (thyroid hormone) and forms hormone receptor complex
- ii. Hormone-receptor complex moves towards the DNA and binds with DNA
- iii. This increases transcription of mRNA
- iv. The mRNA moves out of nucleus and reaches ribosomes and activates them
- v. Activated ribosomes produce large quantities of proteins
- vi. These proteins produce physiological responses in the target cells



HOMEOSTASIS:

Homeostasis refers to the maintenance of constant internal environment of the body.

Physiological factors controlled:

- Core body temperature
- Metabolic wastes (CO₂ and urea)
- Blood PH
- Blood glucose concentration
- Water potential of the blood
- Concentration of respiratory gases in blood

MECHANISM OF ACTION OF HOMEOSTATIC SYSTEM:

- Negative feedback mechanism
- Positive feedback mechanism
- Neural regulation
- Hormonal regulation
- Humoral regulation

NEGATIVE FEEDBACK MECHANISM:

1) REGULATION OF THYROID HORMONE (T3 AND T4) :

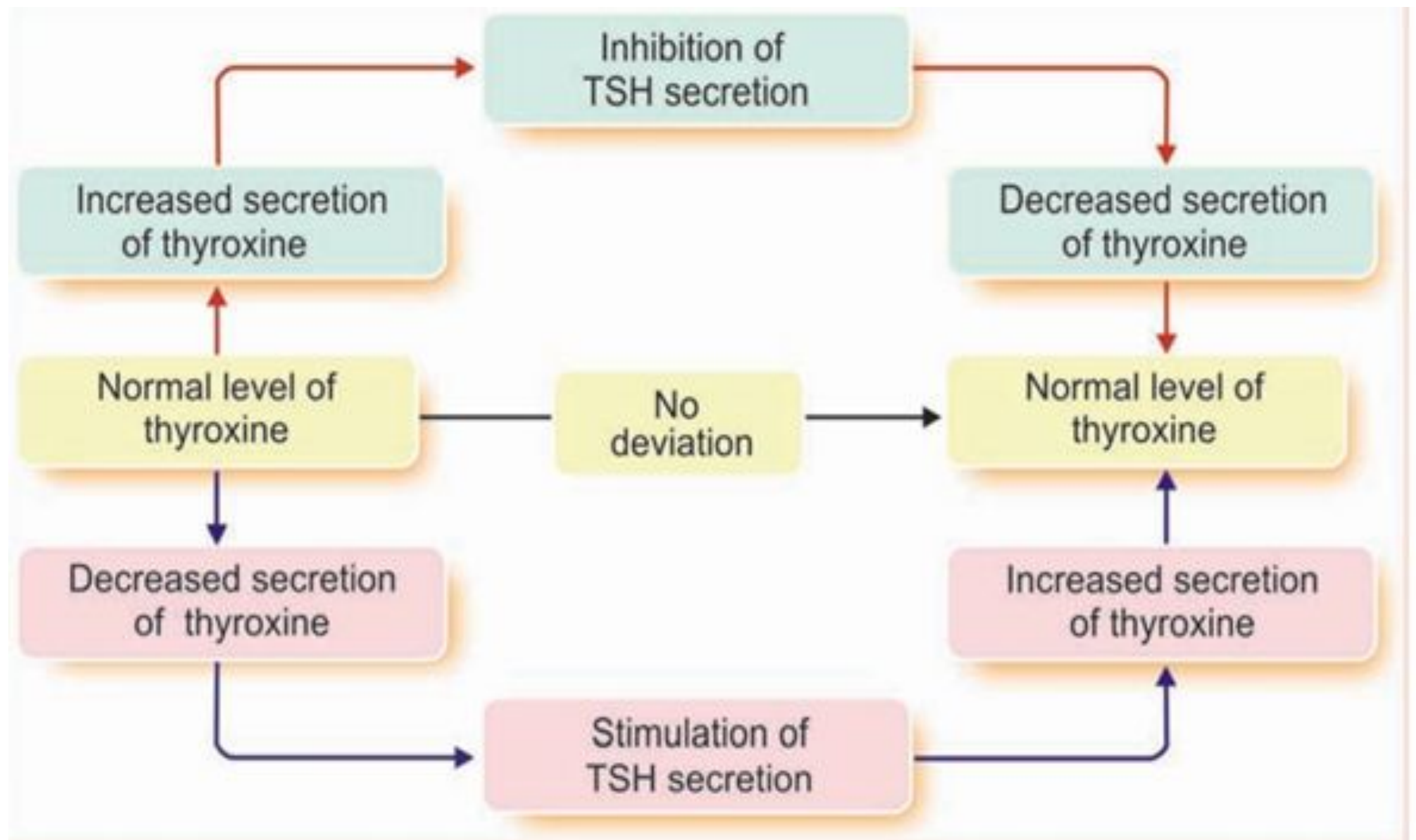
Hormones Involved:

- ❖ TRH(Thyrotropin-Releasing Hormone)-from hypothalamus
- ❖ TSH (Thyroid-stimulating Hormone)-from anterior pituitary
- ❖ T3 (triiodothyronine) and t4(Thyroxine)-from Thyroid gland

Feedback mechanism: Negative feedback loop

Regulation process:

1. Low T3/T4 levels are detected by the hypothalamus.
2. Hypothalamus releases TRH
3. TRH stimulates the anterior pituitary to release TSH
4. TSH stimulates the thyroid gland to release T3 and T4
5. AS T3/T4 levels increase, they inhibit further release of TRH and TSH thus reducing their own production.
6. Maintains stable metabolism, heart rate, and body temperature.



2) REGULATION OF BLOOD GLUCOSE - INSULIN AND GLUCAGON:

Hormone involved:

- Insulin from β -cells of pancreas
- Glucagon from α -cells of pancreas

Feedback mechanism: Negative feedback mechanism

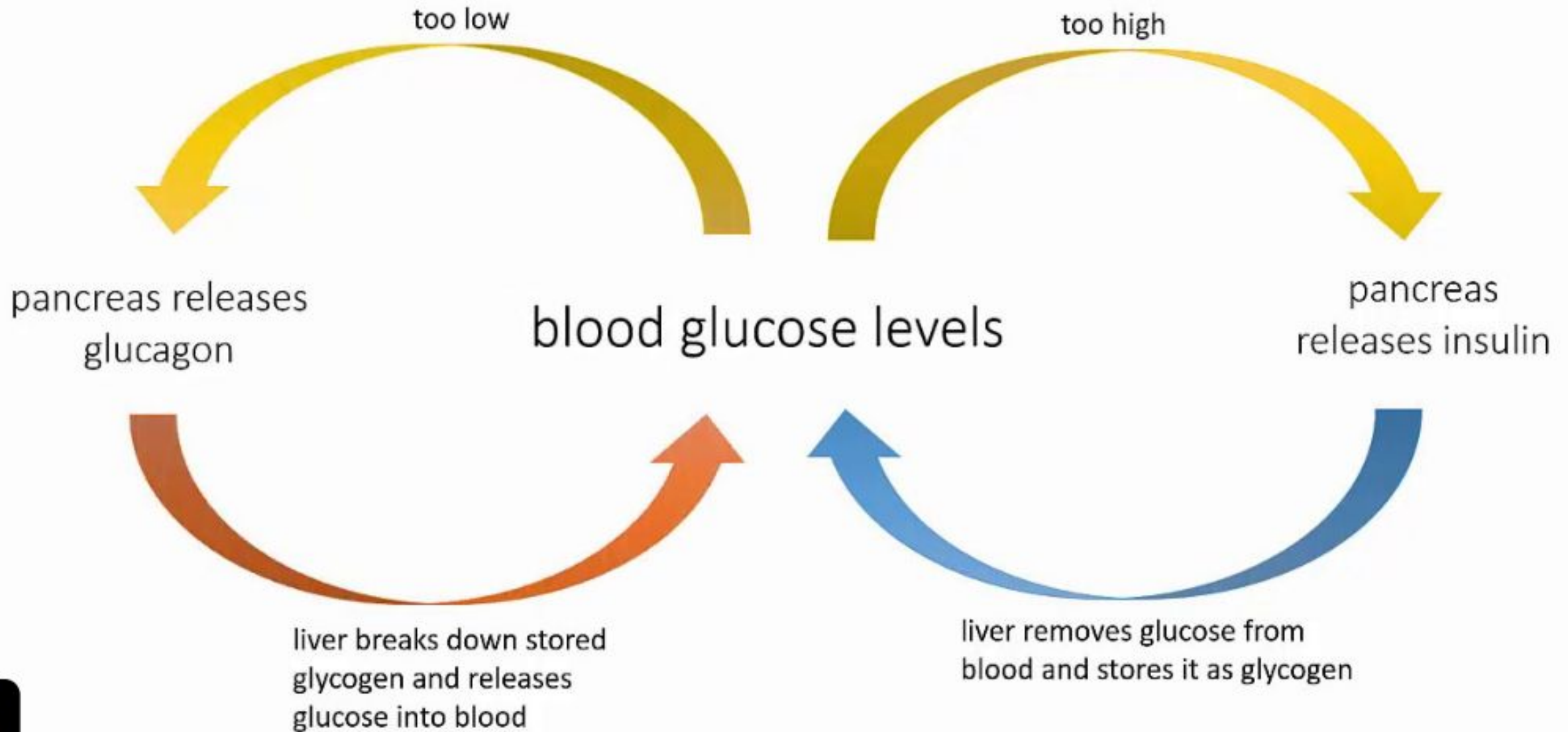
High blood glucose (e.g after a meal):

1. Detected by β -cells in pancreas
2. Insulin is secreted
3. Insulin promote glucose uptake into cells and glycogen synthesis in the liver
4. Blood glucose decreases to normal level
5. Insulin secretion slows down.

Low blood glucose(e.g fasting):

1. Detected by α -cells in pancreas
2. Glucagon is secreted and trigger glycogen breakdown and glucose release from liver
3. Blood glucose rises to normal level
4. Glucagon secretion slows down

NEGATIVE feedback loops: blood sugar



POSITIVE FEEDBACK MECHANISM

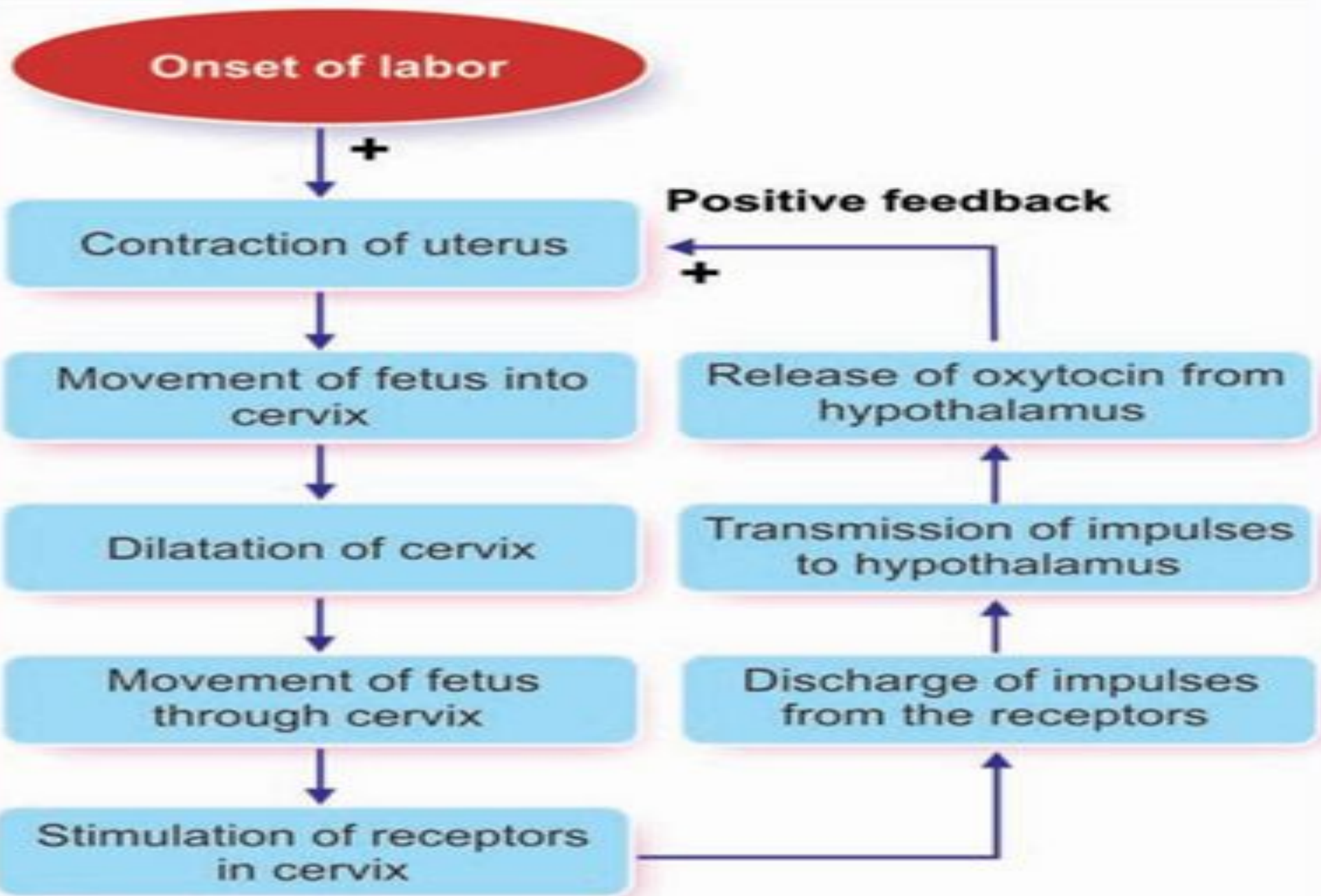
OXYTOCIN IN CHILD BIRTH:

- Oxytocin from posterior pituitary

Feedback mechanism: Positive feedback

Regulation process:

- Baby's head stretches the cervix
- Stretch receptors send signals to hypothalamus
- Hypothalamus signals posterior pituitary to release oxytocin
- Oxytocin causes stronger uterine contractions
- Contractions cause more cervical stretch – more oxytocin
- Loop continues until baby delivered then it stops.

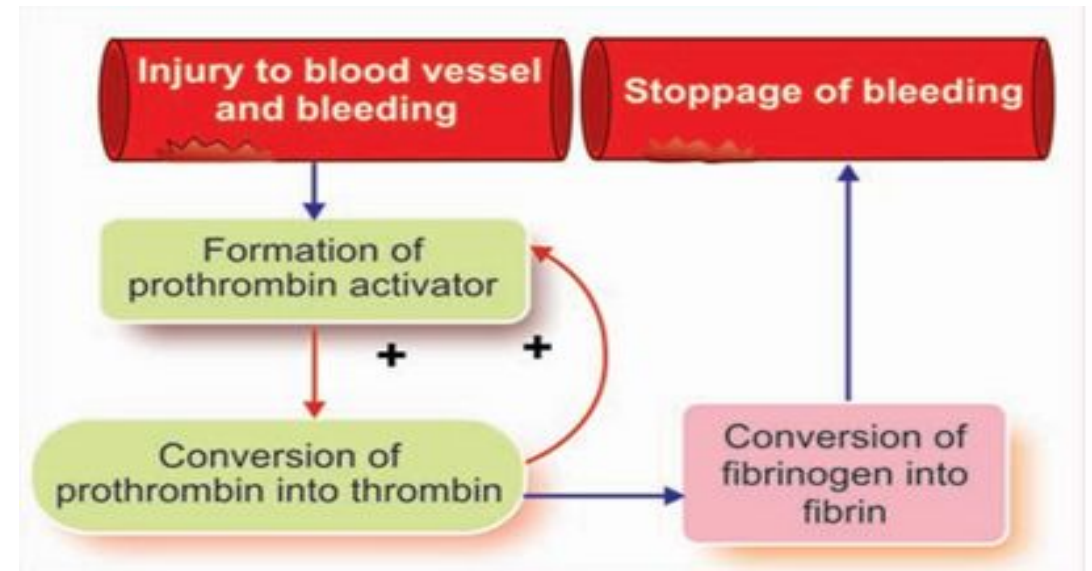


COAGULATION OF BLOOD

Positive feedbacks occurs during the blood clotting. Blood clotting is necessary to arrest bleeding during injury and it occurs in three stages

The three stages are:

- i. Formation of prothrombin activator
- ii. Conversion of prothrombin into thrombin
- iii. Conversion of fibrinogen into fibrin.



The background is a vibrant, abstract composition of organic, flowing shapes in shades of teal, orange, and light pink. Interspersed among these shapes are thin, curved lines and small, scattered dots in various colors, creating a sense of movement and depth. The overall aesthetic is modern and artistic.

Thank You